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ORIGINAL RESEARCH ARTICLE

Contemporary Outcomes and Trends for the Transseptal Mitral Valve-in-Valve Procedure Using Balloon Expandable Transcatheter Valves in the United States

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BACKGROUND: Previous transcatheter valve therapy registry analyses of transcatheter mitral valve in valve (MViV) replacement of degenerated bioprosthesis reported early experience in the United States. Given recent increases in transseptal MViV volumes and introduction of the SAPIEN 3 Ultra valve, it is important to determine contemporary outcomes for patients undergoing transseptal SAPIEN 3/SAPIEN 3 Ultra MViV replacement.

METHODS: The Society of Thoracic Surgeons (STS)/American College of Cardiology Transcatheter Valve Therapy Registry was used to extract data for all patients undergoing transseptal SAPIEN 3/SAPIEN 3 Ultra MViV from 2015 to September 2022. Primary efficacy outcome was 1-year all-cause mortality. Secondary end points included 30-day mortality, functional class, quality of life, and mitral valve performance. Primary safety outcomes were device success and in-hospital complications.

RESULTS: A total of 4243 patients with a mean±SD STS score of 9.2±7.7 underwent transseptal MViV at 455 sites. The rate of Mitral Valve Academic Research Consortium technical (96.6%) success was high, and procedural complications were low. All-cause in-hospital, 30-day, and 1-year mortality rates were 3.2%, 4.3%, and 13.4%, respectively. Significant improvements in New York Heart Association class (New York Heart Association I/II, 18% to 87%) and quality of life (Kansas City Cardiomyopathy Questionnaire score, 38 to 78) were noted at 1 year ($P<0.0001$ for both) after MViV. Upon stratifying by STS scores, it was observed that the low-risk group (STS<4) had a significantly lower in-hospital mortality rate of 0.4%, whereas the intermediate-risk group (STS, 4–8) had an in-hospital mortality rate of 1.9%. From 2015 to 2022, the number of transseptal MViV cases/year increased significantly, whereas procedure times, length of stay, and intensive care unit hours shortened significantly. At the same time, there was a significant trend toward reduced in-hospital ($P=0.0005$), 30-day ($P=0.004$), and 1-year mortality rates ($P=0.01$).

CONCLUSIONS: This multicenter, prospective study reports excellent procedural outcomes, acceptable 1-year mortality rates, and a significant improvement in quality of life for patients undergoing transseptal MViV in the contemporary era. Patients in the low-risk and intermediate-risk STS score categories had significantly better outcomes compared with those in the high-risk category. MViV is a reasonable therapy for the majority of patients with degenerated bioprosthetic mitral valves, who are anatomical candidates.

Key Words: intermediate risk ■ mitral valve in valve ■ MVIV ■ outcomes ■ transseptal

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Clinical Perspective

What Is New?

- This is the largest study to date reporting the contemporary outcomes of transseptal mitral valve in valve (MViV) procedures in the United States. Transseptal MViV was found to be a safe and effective procedure despite its expansion to 455 sites.
- Although MViV is currently approved only for high surgical risk patients, excellent outcomes were noted in low- and intermediate-risk groups as well.
- There was a significant trend toward increasing the volume of procedures and sites over the 8-year study period, with a trend toward reduced mortality rates after Food and Drug Administration approval. There was no volume-outcome relationship; however, there is an initial learning curve associated with 30-day mortality rates.

What Are the Clinical Implications?

- Transseptal MViV should be considered for the majority of anatomically appropriate patients with degenerated bioprosthetic mitral valves after a heart team discussion. Patients on dialysis and those presenting with cardiogenic shock have the highest risk of 1-year mortality after MViV, and shared decision-making is important in these cases.
- There may be an initial learning curve, which warrants appropriate procedural training, education, case selection, and sharing of best practices.
- Future studies should compare the long-term outcomes of transseptal MViV with redo mitral valve surgery in low- and intermediate-risk patients.

Nonstandard Abbreviations and Acronyms

ESRD	end-stage renal disease on dialysis
FDA	Food and Drug Administration
HR	hazard ratio
KCCQ	Kansas City Cardiomyopathy Questionnaire
LVOT	left ventricular outflow obstruction
MR	mitral regurgitation
MS	mitral stenosis
MViV	mitral valve in valve
NT-proBNP	N-terminal pro-B-type natriuretic peptide
NYHA	New York Heart Association
O:E	observed-to-expected
S3	SAPIEN 3
S3U	SAPIEN 3 Ultra
STS	Society of Thoracic Surgeons
TVT	Transcatheter Valve Therapy

In the recent era, operative mortality rates for redo surgical mitral valve replacement range from 6.6% to 11%.[1–4] Transcatheter mitral valve in valve (MViV) offers a less invasive and lower-risk option that was approved by the Food and Drug Administration (FDA) in 2017 for high surgical-risk patients. Two previous studies have reported initial experience of MViV in the United States using the Transcatheter Valve Therapy (TVT) Registry. The first study by Guerrero et al[5] reported the 30-day outcomes in 680 patients from 2013 to 2017. However, 48% of the patients in this study had older-generation SAPIEN and SAPIEN XT transcatheter heart valves that are not in use anymore. Another study reported the outcomes in 1529 patients with SAPIEN 3 (S3) from 2015 to 2019.[6] This study reported that transseptal access was associated with lower 1-year all-cause mortality compared with transapical access. However, this was an early experience with transseptal MViV in the United States. During the past few years, there has been significant growth in transseptal procedures, such as left atrial appendage closure and transcatheter edge-to-edge repair, leading to greater experience and competency in transseptal access among operators. Given the increase in volume of transseptal MViV procedures, introduction of SAPIEN 3 Ultra (S3U), mandatory use of computed tomography scan planning, and left ventricular outflow obstruction (LVOT) risk mitigation techniques, an updated re-evaluation of the transseptal outcomes is warranted.

This TVT Registry–based analysis aims to update previous analyses and report the contemporary clinical outcomes of patients who underwent transcatheter transseptal MViV procedures using S3/S3U valves in the United States. The secondary objectives were to assess the temporal trends from 2015 to 2022, the volume-outcome relationship, the learning curve, and predictors of all-cause mortality.

METHODS

Study Population

This study included all patients who underwent a transcatheter transseptal MViV procedure in the United States from August 2015 to September 2022 and were reported to the national Society of Thoracic Surgeons (STS)/American College of Cardiology TVT Registry. The data were maintained by the TVT Registry and are frequently audited for accuracy.[7] The TVT Registry has been approved by a central institutional review board (Advarra) and granted a waiver of informed consent by the Duke University School of Medicine institutional review board under the Common Rule 45 CFR 46.3. This article conforms with the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) cohort guidelines.

The main inclusion criterion of this study was the use of transseptal access for MViV. Of the total 4666 patients undergoing MViV, 4243 patients had transseptal access. All patients with transapical access (n=314), direct left atrium (n=11), or

other access routes (n=98) were excluded. All the echocardiographic data including grade of mitral regurgitation (MR) and mean gradient were site-reported. Severe mitral stenosis (MS) was defined as a mean gradient of >10 mm Hg or mitral valve area <1.5 cm^2. The authors declare that all supporting data are available within the article (and its Supplemental Material).

Transseptal MViV

The procedure has been described previously.[8] In short, transseptal MViV is performed through transfemoral venous access followed by transseptal puncture and atrial septum dilation. Then the transcatheter heart valve is delivered over a stiff wire and deployed inside the previous bioprosthetic mitral valve. Sites used their discretion during various steps of the procedure according to local expertise and experience. Atrial septal defect closure was performed according to operator preference.

Procedural and Clinical Outcomes

The primary outcome of interest was 1-year all-cause mortality. Secondary outcomes included in-hospital, 30-day, and 1-year mortality, cardiac mortality, stroke, LVOT obstruction, mitral valve reintervention, atrial septal defect closure, major vascular complications, New York Heart Association (NYHA) class, and quality of life defined by the 12-item Kansas City Cardiomyopathy Questionnaire (KCCQ).

Implant success was defined as successful delivery and deployment of the transcatheter heart valve. Technical success was defined using the Mitral Valve Academic Research Consortium criteria as: (1) absence of procedural mortality; (2) successful access, delivery, and retrieval of the device delivery system; (3) successful deployment and correct positioning of the first intended device; and (4) freedom from emergency surgery or reintervention related to the device or access procedure.[9] Procedural complications included conversion to open heart surgery, device embolization, device migration, or cardiac perforation as defined by the TVT Registry guidelines.[10]

One-year follow-up was initially determined using the TVT Registry for all patients from 2015 to 2022 (Figure S1). This resulted in 72.9% complete follow-up data, as 737 patients were not due for their follow-up at the time of this study. This was supplemented using the US Centers for Medicare & Medicaid Services claims data for patients ≥65 years age, which provided an additional 493 patients at risk of 1-year mortality. Thus, 86.9% complete follow-up was available at 1 year. The Centers for Medicare & Medicaid Services data were recently made available and provide follow-up only until December 2020. A probabilistic matching method was used, and patients from the TVT Registry were matched with Centers for Medicare & Medicaid Services patients on the basis of the exact MViV procedure date, birth date, and sex. Patients with multiple matches were removed from the analysis (only one unique case was used). Observed-to-expected (O:E) ratio is reported where the expected mortality rate is based on the STS score of surgical mortality.

Statistical Analysis

Continuous variables were presented as mean (SD) or median (interquartile range) and were compared between groups using the 2-sample *t* test or Wilcoxon rank sum test. Categorical variables were presented as frequencies and percentages and were compared using the χ^2 or Fisher exact test. The 30-day and 1-year adverse event rates were based on Kaplan-Meier estimates, and all comparisons were made using the log-rank test. For the scatter plot of the volume-outcome relationship after MViV, unadjusted and adjusted Cox regression models with restricted cubic splines were used to model the relationship between 30-day mortality and total site volume. We then stratified the site volume into tertiles on the basis of the number of MViV cases performed, resulting in low-, intermediate-, and high-volume sites with nearly the same number of patients in each group. Kaplan-Meier and Cox regression–adjusted analyses were then performed to assess the association of tertiles with 30-day all-cause mortality. For the learning curve analysis, the relationship between case sequence across all sites and 30-day mortality was modeled using a Cox regression with restricted cubic splines. The relationship was illustrated in terms of the logarithm of the hazard ratio, allowing for the interpretation of the magnitude and direction of the association. Each k-th case sequence from all sites was used to model the relationship (k=1,…, n; where n is the largest number of cases performed among sites). For example, in the case sequence of 2, the results were compiled for the second case among all sites. For the temporal trend analysis, the study was divided into 3 time periods: pre-FDA approval (n=257; August 2015 to June 5, 2017), post-FDA (n=1616; June 6, 2017, to December 2019), and contemporary (n=2370; January 2020 to September 2022).

Multivariable analyses were performed to identify independent predictors of 30-day and 1-year mortality. Baseline comorbidities with a *P* value <0.1 in the univariable analysis were entered in the multivariable model. A Cox regression model with the lowest Akaike Information Criterion was selected for its optimal balance of model complexity. For the 30-day multivariable analysis, 17 covariates were used for adjustment to prevent model overfitting: age (every 10 years), hemoglobin, glomerular filtration rate, end-stage renal disease on dialysis (ESRD), immunocompromised (definition in Supplemental Material), cardiogenic shock within 24 hours, cardiac arrest within 24 hours, left ventricular ejection fraction, peripheral arterial disease, chronic lung disease, home oxygen, moderate-severe or severe MR, severe tricuspid regurgitation, elective versus nonelective (urgent, emergency, or salvage) procedure, mechanical support at the start of procedure, NYHA III/IV, and KCCQ score. For the 1-year multivariable analysis, 30 covariates were used for adjustment: age (every 10 years), hemoglobin, glomerular filtration rate, permanent pacemaker, previous implantable cardioverter defibrillator, previous percutaneous coronary intervention, previous coronary artery bypass surgery, peripheral arterial disease, currently on dialysis, hostile chest, immunocompromised, heart failure hospitalization within past year, porcelain aorta, atrial fibrillation/flutter, chronic lung disease, previous myocardial infarction, cardiogenic shock within 24 hours, cardiac arrest within 24 hours, moderate-severe or severe MR, severe aortic regurgitation, aortic stenosis, severe tricuspid regurgitation, previous aortic valve procedure, diabetes, home oxygen, severe bioprosthetic MS, elective versus nonelective procedure, mechanical support at the start of procedure, NYHA III/IV, and KCCQ score. Proportional hazards assumption

was confirmed through testing based on Kolmogorov-type supremum test. The restricted spline curves were created using R, version 4.3.1. All other statistical analyses were performed using SAS, version 9.4 (SAS Institute), and statistical significance was set at a 2-sided $P<0.05$ without multiplicity adjustment.

RESULTS

Baseline Characteristics

A total of 4243 patients underwent transseptal MViV at 455 sites in the United States between August 2015 and September 2022. Baseline demographic and echocardiographic characteristics are presented in Table 1. Mean age was 73.0±11.5 years, and 58.8% were female. Most patients were White; 10% were Black, and 6.27% were of Hispanic or Latino ethnicity. This was a high-risk population with a mean (SD) STS score of 9.2 (7.7) and many extreme risk factors such as cardiogenic shock in 3.9%, mechanical support devices in 2.9%, preprocedural inotropes in 8.9%, ESRD in 5%, home oxygen in 13.7%, and need for urgent, emergent, or salvage procedure in 25.2%. NYHA class III/IV symptoms were present in 82.4% of the patients, and BNP (B-type natriuretic peptide)/NT-proBNP (N-terminal pro-B-type natriuretic peptide) levels were severely elevated. Baseline mean pulmonary capillary wedge pressure was severely elevated with a mean of 26.2±8.4 mm Hg (Table 1). Severe bioprosthetic MS was present in 57.1%, moderate-severe or severe MR in 18.5%, and mixed severe MS and moderate-severe MR were present in 24.4% of the patients. Concomitant severe tricuspid regurgitation was present in 20.2% of the patients.

Procedural Characteristics

Implant success was achieved in 97.1% of the patients. Mitral Valve Academic Research Consortium technical success was achieved in 96.6% of the patients. Reasons for technically unsuccessful procedure included death (n=34), mitral valve reintervention (n=4), aborted procedure (n=12), conversion to surgery (n=11), and unsuccessful device implantation (n=84). General anesthesia was used in 94.2% of patients, whereas moderate sedation was used in 5.5%. S3U was introduced in 2018 and was used in 22.3% of the patients, whereas S3 was used in others (Table 2). The rate of procedural complications was low: cardiac perforation, 1.0%; conversion to open heart surgery, 0.7%; device embolization, 0.18%; device migration, 0.3%. Postprocedure median length of stay was 2 days, and median intensive care unit stay was 23 hours. A total of 86.3% of patients were discharged home. Of all the patients, 81.8% were discharged on anticoagulation, and 73.8% were discharged on an antiplatelet agent. Table S1 shows the discharge medications stratified by history of atrial fibrillation.

Table 1. Baseline Characteristics of Patients Undergoing Transseptal MViV

Characteristics	% (n/N)
Baseline characteristics	
Age, y	73.0±11.5
Female	58.8% (2494/4243)
Race and ethnicity	
White	83.0% (3522/4243)
Black	10.0% (425/4243)
Others*	6.9% (296/4243)
STS score	9.2±7.74 (3961)
Previous ICD	11.2% (475/4232)
Previous PCI	16.4% (695/4237)
Previous CABG	30.6% (1296/4236)
Previous aortic valve procedure	24.1% (1021/4241)
MVR type	
Stented	89.7% (2184/2434)
Stentless	10.3% (250/2434)
Previous stroke	17.6% (748/4242)
ESRD	4.9% (211/4237)
Immunocompromised	7.5% (305/4095)
Heart failure hospitalization within past year	57.8% (2096/3627)
Atrial fibrillation/flutter	72.4% (3069/4242)
Chronic lung disease	39.9% (1686/4217)
Home oxygen	13.7% (582/4240)
Diabetes	26.7% (1132/4238)
Endocarditis	14.3% (608/4239)
NYHA within 2 weeks	
I	1.9% (81/4174)
II	15.7% (654/4174)
III	55.6% (2323/4174)
IV	26.7% (1116/4174)
Cardiogenic shock within 24 hours	3.9% (166/4241)
Cardiac arrest within 24 hours	0.4% (19/4240)
Preprocedure use of inotropes	8.9% (374/4215)
BNP	789.4 ± 1025.7 (1813)
NT-proBNP	5354.1 ± 8219.1 (1414)
Heart team reason for procedure†	
Inoperable/extreme	17.7% (745/4206)
High	63.2% (2657/4206)
Intermediate	14.8% (621/4206)
Low	4.4% (183/4206)
Hemodynamics (preprocedure)	
Mean PCWP, mm Hg	26.2±8.5 (2151)
Mean PAP, mm Hg	40.6±12.3 (2245)
Mean RA, mm Hg	12.1±6.6 (2401)
Echocardiographic characteristics	
Mitral mean gradient, mm Hg	13.1±6.0 (4045)
Left ventricular ejection fraction, %	55.0±11.8 (4198)

(*Continued*)

Table 1. Continued

Characteristics	% (n/N)
Mitral regurgitation	
None/trace/mild	46.6% (1958/4204)
Moderate	12.7% (534/4204)
Moderate-severe	7.1% (299/4204)
Severe	33.6% (1413/4204)
Paravalvular leak (≥moderate)	11.8% (219/1862)
Tricuspid regurgitation (severe)	20.2% (852/4226)

Data presented as % of patients (No./total No.) for categorical variables and mean±SD for continuous variables. CABG indicates coronary artery bypass graft surgery; ESRD, end-stage renal disease on dialysis; ICD, implantable cardioverter defibrillator; MVR, mitral valve replacement; PAP, pulmonary artery pressure; PCI, percutaneous coronary intervention; PCWP, pulmonary capillary wedge pressure; RA, right atrial pressure; and STS, Society of Thoracic Surgeons.

*Others include Asian (2.0%), American Indian/Alaskan Native (0.68%), Native Hawaiian/Pacific Islander (0.35%), and unknown (3.9%).

†Definition in Supplemental Material.

In-Hospital, 30-Day, and 1-Year Clinical Outcomes

In this high-risk cohort, in-hospital all-cause and cardiac mortality rates were 3.2% and 1.8%, respectively. LVOT obstruction was low at 0.4%, and atrial septal defect closure was performed on 8.2% of patients (Table 3). The rate of 30-day all-cause mortality was 4.3% with an O:E ratio of 0.48. At 1 year, all-cause mortality was reported in 13.4% of the patients, of whom cardiac mortality was reported in 4.5%. At 1 year, mitral valve reintervention was reported in 1.4% of patients. A total of 31.4% patients had readmissions during the first year, of which 10.2% were related to heart failure.

Predictors of Mortality

Univariable predictors of 1-year mortality are presented in Table S2. Figure 1A shows the multivariable predictors of 30-day mortality after MViV. Figure 1B shows the multivariable predictors of all-cause mortality at 1 year. Preprocedure ESRD (hazard ratio [HR], 2.5 [95% CI, 1.5–3.8]) and cardiogenic shock (HR, 2.5 [95% CI, 1.5–4.1]) before the procedure were the strongest risk factors for 1-year mortality after MViV. Other independent predictors of 1-year mortality included older age, lower baseline KCCQ score, lower glomerular filtration rate, lower hemoglobin, immunocompromised status, prior aortic valve procedure, prior percutaneous coronary intervention, home oxygen, and peripheral arterial disease.

We further stratified the patients by surgical risk using the mean STS score. Patients in the low-risk group (mean STS score, 2.7) were younger and had a very low in-hospital mortality rate of 0.4% (n=831), with O:E of 0.28 (Table S3). One-year mortality rate was also significantly lower compared with intermediate-risk and high-risk groups (4.7% versus 10.1% versus 20.1%; $P \leq 0.0001$). In-hospital mortality in the intermediate-risk group (mean STS score, 5.9) was 1.9%. Adjusted 1-year mortality was significantly lower in the low-risk (HR, 0.19 [95% CI, 0.13–0.29]; $P \leq 0.0001$) and intermediate-risk (HR, 0.45 [95% CI, 0.36–0.56]; $P \leq 0.0001$) groups compared with the high-risk group (Figure 2).

Echocardiographic and Functional Outcomes

Figure 3 presents the echocardiographic outcomes after MViV. There was a significant and durable improvement in MR after MViV; 99.2% of patients had no or mild MR at discharge, 30 days, and 1 year (Figure 3A). The mean gradients after MViV at discharge and 1 year were 6.7 mm Hg and 7.4 mm Hg, respectively (Figure 3B). The mean gradients at discharge for valve sizes 23 mm and 26 mm were 8.1±3.2 mm Hg and 7.0±2.7 mm Hg, respectively.

Figure S2 shows the overall and paired analysis of change in NYHA class from baseline to 1 year. For the paired analysis, the majority (81.8%) of the patients were in NYHA class III/IV at baseline. Significant improvement in NYHA class was noted at 30 days and 1 year, when 85.3% of patients were in NYHA class I/II at 1 year. Quality of life as determined by KCCQ score also improved significantly from a baseline score of 36.8 to 73.0 at 30 days and 78.0 at 1 year (Figure S3).

Temporal Trends

Figure 4 shows that there has been a significant upward trend in the number of transseptal MViV cases and sites since 2015. The highest growth occurred right after FDA approval in 2017 with a consistent rise after that. Median (interquartile range) number of cases per site was 5 (2 to 12). Figure S4 shows the distribution of sites by the number of cases.

Tables S4 through S6 present the patient, echocardiographic, and procedural characteristics during the 3 time periods (as defined above). The number of sites performing transseptal MViV increased from 86 in the pre-FDA group (August 2015 to June 5, 2017) to 306 after FDA approval (June 6, 2017, to December 2019) to 420 in the contemporary era (January 2020 to September 2022). Overall, there were no major differences in patient characteristics. Temporally, there was an increase in the frequency of intermediate-risk and low-risk patients as defined by the STS score, nonelective procedures, and moderate sedation, whereas frequency of aborted procedures ($P<0.0001$), median procedure times ($P<0.0001$), median hospital length of stay ($P \leq 0.0001$), and median intensive care unit hours ($P \leq 0.0001$) decreased significantly (Table S4).

In-hospital all-cause mortality was highest in the pre-FDA approval era (6.6%; $P=0.006$) and then plateaued after FDA approval and in the contemporary era (2.9% and 3.0%; Table S7). Similar trends were observed for 1-year mortality (Figure 5). After adjusting for baseline variables, patients in the pre-FDA approval era had a significantly higher risk of mortality compared with the

Table 2. Procedural Characteristics and Complications in Patients Undergoing Transseptal MViV

	% (n/N)
Procedural characteristics	
Device implant success	97.1% (4119/4243)
MVARC technical success	96.6% (4098/4243)
Procedure status	
Elective	74.8% (3173/4240)
Urgent	23.3% (986/4240)
Emergency/salvage	1.9% (81/4240)
Type of anesthesia	
General anesthesia	94.2% (3994/4238)
Moderate sedation	5.5% (233/4238)
Total procedure time, min	95.0 [72.00, 127.00]
Valve size	
20 mm	0.1% (4/4241)
23 mm	6.3% (269/4241)
26 mm	44.4% (1882/4241)
29 mm	49.2% (2086/4241)
SAPIEN 3	77.6% (3293/4243)
SAPIEN 3 Ultra	22.3% (948/4243)
SAPIEN 3 Ultra RESILIA	0.1% (2/4243)
Procedural complications	
Device embolization	0.2% (8/4243)
Device migration	0.3% (12/4243)
Conversion to open heart surgery	0.6% (26/4243)
Mechanical assist sevice*	2.9% (123/4243)
Transseptal complication	0.6% (27/4243)
Cardiac perforation–in hospital	0.9% (36/4243)
Index hospitalization	
Length of stay, days	2.0 [1.0, 4.0]
ICU length of stay, hours	23.0 [0.0, 36.9]
Discharge location	
Home	86.3% (3663/4243)
Hospice	0.2% (10/4243)
Nursing home	3.5% (150/4243)
Extended care/TCU/rehabilitation	5.7% (242/4243)

Data presented as % of patients (No./total No.) or median [interquartile range]. MVARC indicates Mitral Valve Academic Research Consortium.

*Mechanical assist device includes IABP, Impella, ECMO, or bypass.

contemporary era (adjusted HR, 1.43 [95% CI, 1.01–2.02]; *P*=0.04). There was no significant difference in the rate of 1-year mortality between the post-FDA approval and contemporary eras. The risk of in-hospital LVOT obstruction reduced significantly over the study time periods (1.6% versus 0.6% versus 0.2%; *P*=0.004).

Volume-Outcome Relationship and Learning Curve Assessment

Figure 6A shows the spline curves for total site volume and 30-day mortality. The adjusted analysis does not suggest any significant association. Further analysis using tertiles of site volume also showed no significant difference in 30-day mortality between the tertiles (Figure 6B). For learning curve assessment, spline curves for case sequence analysis for each site suggested a change at 10 cases (Figure S5A). The 30-day mortality rate was significantly higher for the first 10 cases (5.2%) compared with the next 10 cases (3.0%; adjusted HR, 0.64 [95% CI, 0.42–0.9]; *P*=0.04; Figure S5B).

Table 3. In-Hospital, 30-Day, and 1-Year Clinical Outcomes After Transseptal MViV

Outcomes	In-hospital	30-day*	1-year*
All-cause mortality	3.2% (136/4243)	4.2% (173)	13.4% (417)
Cardiac death	1.8% (76/4243)	2.1% (85)	4.5% (1466)
Mitral valve reintervention	0.4% (16/4243)	0.5% (19)	1.4% (41)
Major vascular complication	1.2% (52/4243)	1.4% (60)	1.72% (67)
New requirement for dialysis	1.1% (46/4243)	1.2% (49)	1.8% (64)
Periprocedural MI	0.12% (5/4243)	0.14% (6)	0.14% (6)
New pacemaker or ICD	0.94% (38/4056)	1.2% (47)	2.6% (79)
Stroke	0.94% (40/4243)	1.3% (52)	2.7% (87)
TIA	0.12% (5/4243)	0.2% (9)	0.5% (16)
LVOT obstruction	0.4% (18/4243)	0.4% (18)	0.4% (18)
Device thrombosis	N/A	0.22% (9)	0.41% (14)
ASD closure	8.2% (346/4243)	8.2% (349)	8.9% (365)
Readmissions (all)	0.2% (10/4243)	9.0% (352)	31.4% (923)
Readmissions, cardiac (not HF)†	0.09% (4/4243)	1.5% (58)	7.3% (202)
Readmissions, heart failure†	0.02% (1/4243)	2.3% (90)	10.2% (292)

Data presented as % of patients (No./total No.) for in-hospital outcomes and as % (No. of events) for 30-day and 1-year outcomes. ASD indicates atrial septal defect; HF, heart failure; ICD, implantable cardioverter defibrillator; LVOT, left ventricular outflow obstruction; MI, myocardial infarction; and TIA, transient ischemic attack.

*Event rates were calculated by Kaplan-Meier method.

†Definition of readmissions (cardiac; not heart failure) and readmissions (heart failure) are included in the Supplemental Material.

DISCUSSION

This large, prospective, observational study reports that transseptal MViV with S3 and S3U was associated with excellent procedural outcomes, low complication rates, significant improvement in valve performance with no or mild MR in 99.2%, and an acceptable mortality rate in a high-risk population. The mortality rate was largely influenced by the patients' risk factors and the presence of other medical conditions, as demonstrated by the fact that the low-risk group had an in-hospital mortality rate of only 0.4%. Transseptal MViV led to a marked +41-point improvement in KCCQ score and significant improvement in NYHA class at 1 year. Since FDA approval in 2017, there was a significant increase in the

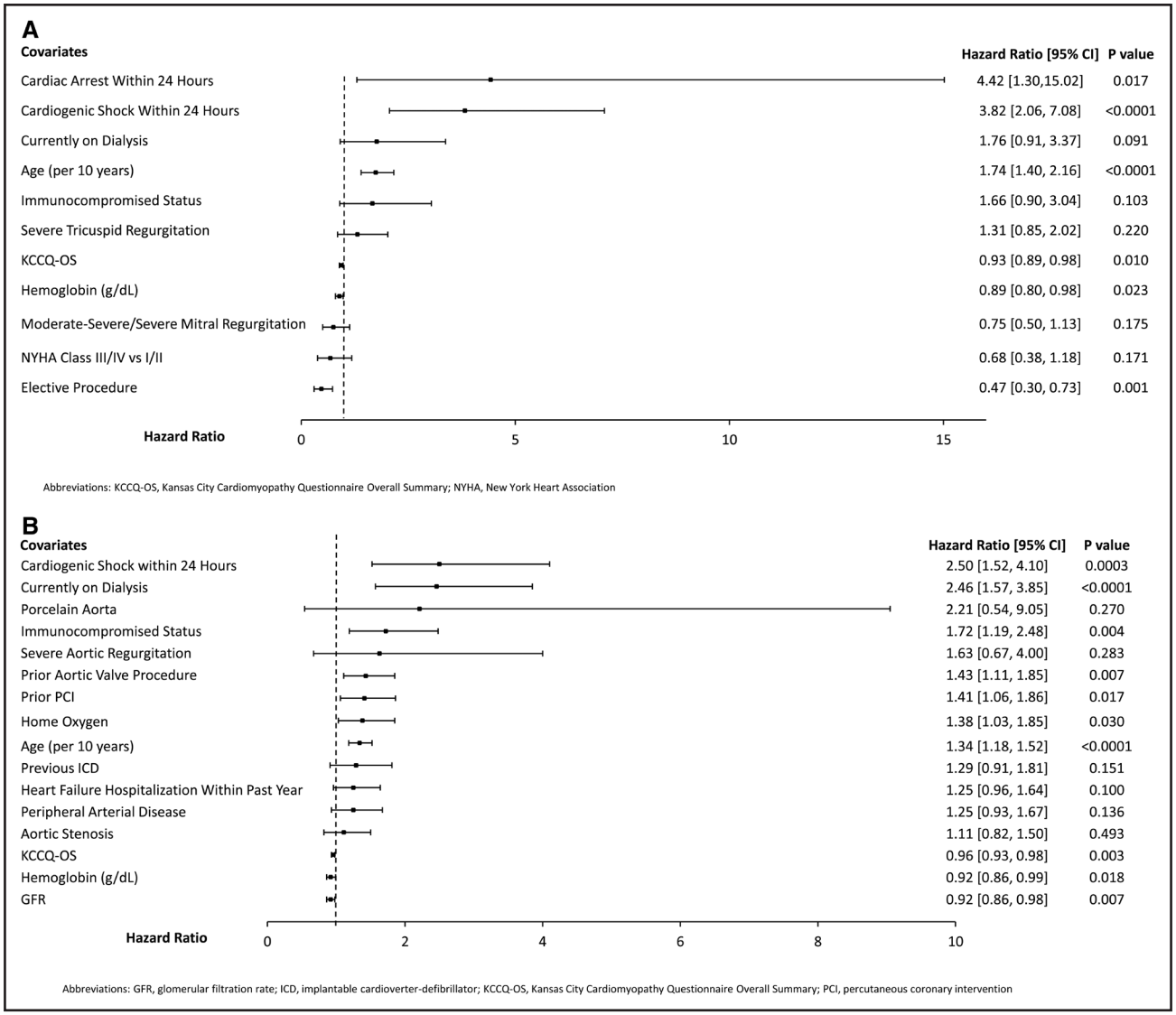

Figure 1. Multivariate predictors of all-cause mortality after MViV.
A, Multivariable predictors of all-cause mortality at 30 days after MViV. Seventeen covariates: age (every 10 years), hemoglobin, GFR, currently on dialysis, immunocompromised, cardiogenic shock within 24 hours, cardiac arrest within 24 hours, LVEF, NYHA III/IV, KCCQ score, peripheral arterial disease, chronic lung disease, home oxygen, moderate-severe or severe mitral regurgitation, severe tricuspid regurgitation, elective procedure status, and mechanical support at start of procedure. **B**, Multivariable predictors of all-cause mortality rates at 1 year after MViV. Thirty covariates: permanent pacemaker, previous implantable cardioverter defibrillator (ICD), previous percutaneous coronary intervention (PCI), previous coronary artery bypass surgery, peripheral arterial disease, currently on dialysis, hostile chest, immunocompromised, heart failure hospitalization within past year, porcelain aorta, atrial fibrillation/flutter, chronic lung disease, previous myocardial infarction, cardiogenic shock within 24 hours, cardiac arrest within 24 hours, moderate-severe or greater mitral regurgitation, severe aortic regurgitation, aortic stenosis, severe tricuspid regurgitation, hemoglobin, glomerular filtration rate (GFR), age (every 10 years), previous aortic valve procedure, diabetes, home oxygen, mitral stenosis (gradient ≥10 mm Hg or valve area <1.5 mm), elective procedure status, mechanical support at start of procedure, NYHA III/IV, and KCCQ score.

volume of transseptal MViV procedures and sites along with improvement in procedural and 1-year outcomes. Although the median volume per site was low, there was no association between overall site volume and 30-day mortality after transseptal MViV. On the basis of STS score, low-risk and intermediate-risk patients have significantly better outcomes compared with the high-risk group. Overall, these data suggest that transseptal MViV is safe and effective and should be considered for the majority of patients with degenerated bioprosthetic mitral valves, who are anatomical candidates.

This is the largest study to date reporting the outcomes of transseptal MViV for bioprosthetic mitral valve degeneration. In this unselected, high-risk, real-world population with many extreme risk factors such as ESRD, cardiogenic shock, home oxygen, and emergency procedures, the in-hospital mortality rate was 3.2% and the 1-year mortality was 13.4%, with an O:E

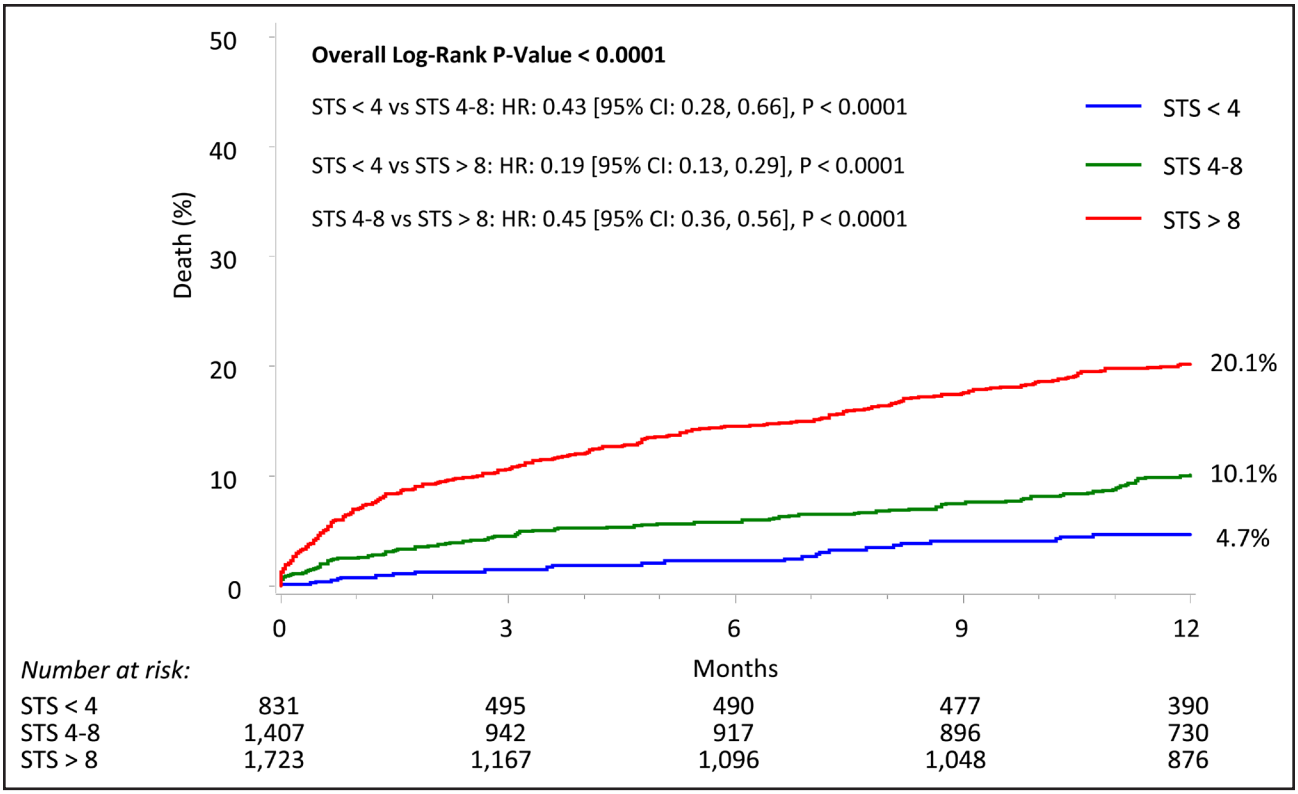

Figure 2. One-year all-cause mortality rates by low-risk, intermediate-risk, and high-risk STS scores.
Low, STS <4; intermediate, STS 4 to 8; high, STS >8. STS indicates Society of Thoracic Surgeons score.

ratio of 0.48. This is lower than redo mitral valve surgery, for which operative mortality rate ranges from 6.6% to 11.1% and O:E is 0.72 to 0.82 in the recent era.[1–4] The 1-year mortality rate is also significantly lower than high surgical-risk patients undergoing transcatheter mitral valve replacement for native MR, which ranges from 25% to 30%[11]; transcatheter edge-to-edge repair for primary MR, with a mortality rate of 15.4%[12]; secondary MR, with mortality risk of 22.2%[13]; valve in mitral annular calcification, with a 1-year mortality rate up to 43%[14]; or mitral valve in ring, with a 1-year mortality rate of 23%.[15] The superior outcomes observed with MViV could be related to its high success rate (Mitral Valve Academic Research Consortium technical success, 96.6%), excellent safety profile, reproducibility of the procedure, and patients selected for the procedure. Furthermore, a remarkable improvement in the KCCQ score from baseline to 1 year emphasizes early, sustained, and clinically meaningful improvements in quality of life, much higher than previously reported.

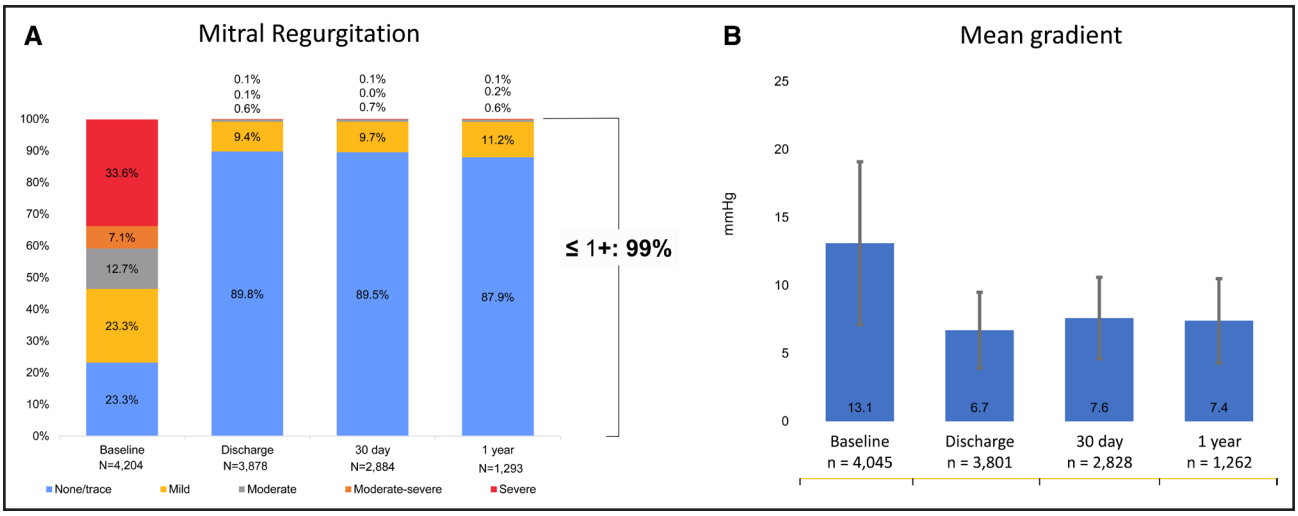

Figure 3. Valve performance at discharge, 30 days, and 1 year.

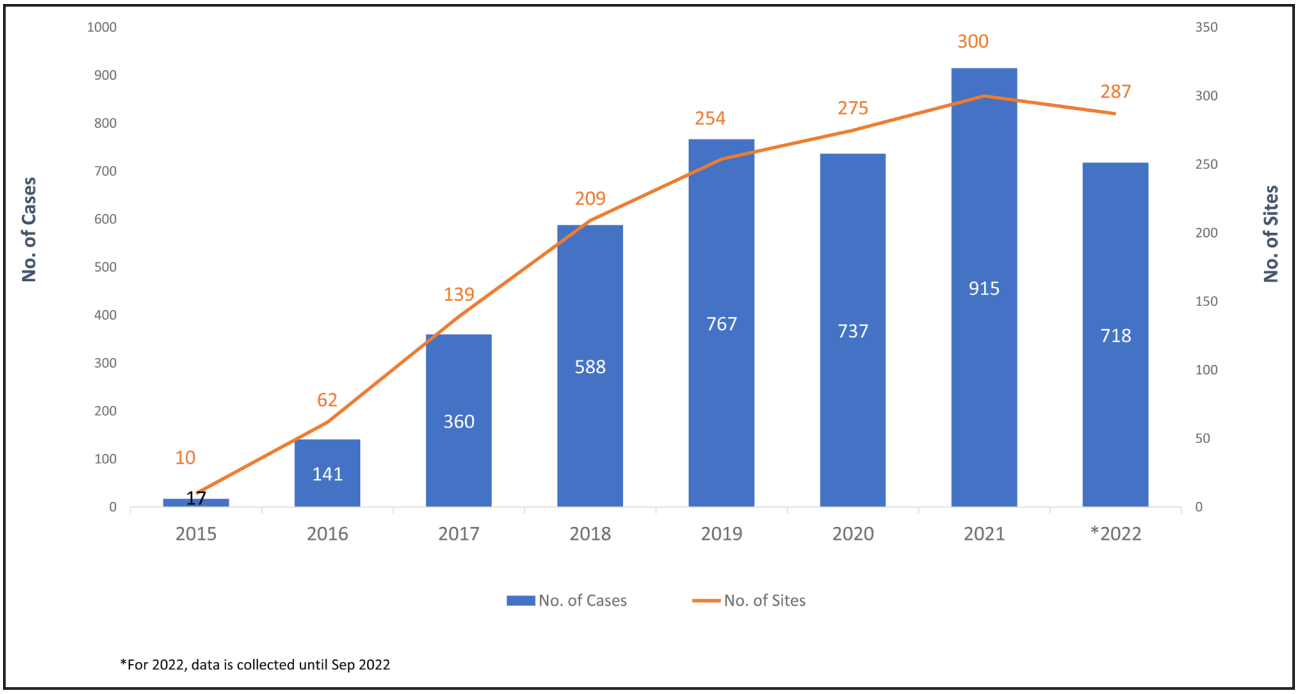

Figure 4. Temporal trends in MViV cases and sites in the United States.

The current analysis provides a substantial update compared with the previously published TVT Registry studies[5,6] and other small single or multicenter studies.[16] The previous studies reported early experience after the introduction of MViV in the United States, were limited to older-generation transcatheter heart valves or S3 only, and included transapical access, which is associated with significantly increased mortality rates compared with transseptal access. We focused on transseptal access only because it is the most common and safe approach and extended the duration to September 2022, resulting in a 3× larger study population of transseptal MViV

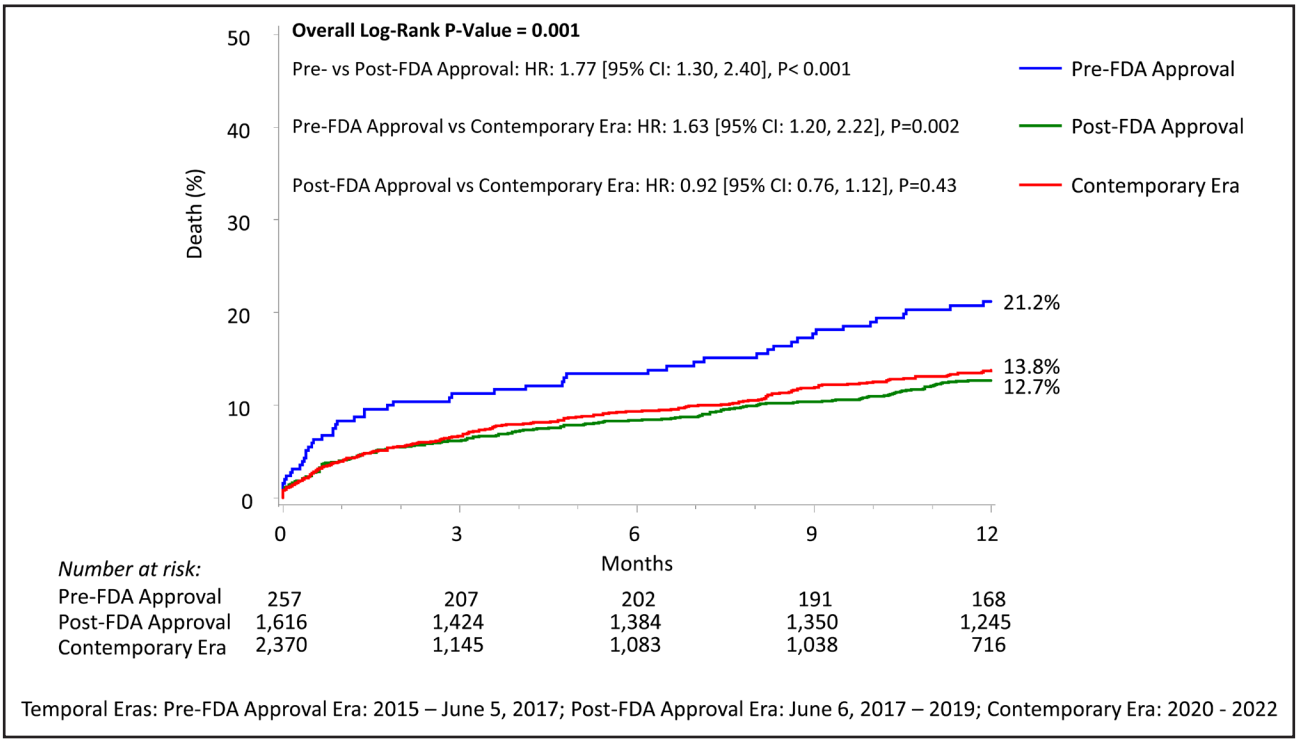

Figure 5. One-year all-cause mortality by time periods.
FDA indicates Food and Drug Administration.

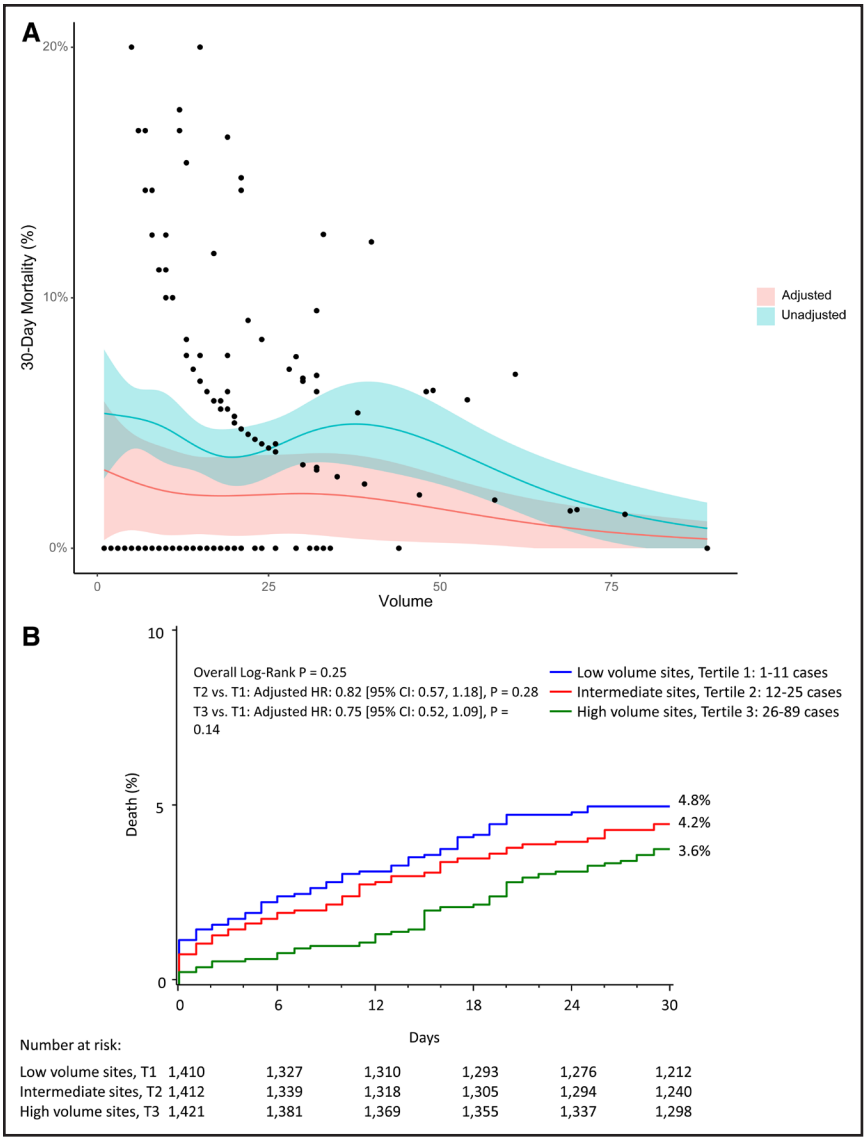

Figure 6. Volume outcome relationship after MViV.
A, Scatter plot of volume-outcome relationship after MViV. **B**, One-year all-cause mortality rates by low-, intermediate-, and high-volume site tertiles.

than the previous studies. With longer study duration and follow-up, the reported outcomes are more robust. In addition, we report the temporal trends in the United States, combined outcomes with S3U, volume-outcome relationship, and learning curve analysis for the first time in patients undergoing MViV. We found that the volume of procedures and sites increased significantly, whereas the procedure time, length of stay, and intensive care unit hours decreased over time. FDA approval coincided with significant improvement in outcomes, including a reduction in the rates of in-hospital LVOT obstruction, in-hospital mortality, and 1-year mortality. FDA approval of a procedure establishes its safety and efficacy, thus allowing its expansion to newer sites and operators. The improved outcomes are most likely secondary to overcoming the learning curve at several sites, the use of computed tomography scans for planning and execution, extensive procedural training, availability and use of proctors for new operators and sites, and growing experience with transseptal access.

This is the first study to explore the relationship of volume and outcomes after MViV. The median volume per site was low at 5, and only 53 sites performed >20 cases during the study period. When we assessed the relationship between site volume and 30-day mortality, there was no significant difference between the lowest and highest tertiles. This could be related to the careful selection of cases by new and low-volume centers or the training of more structural operators. However, there is a learning curve that was noticed per site. In the case sequence analysis, we noticed a significant decrease in 30-day mortality after the first 10 cases at a site. The registry does not collect individual operator data, so it cannot be determined whether multiple operators were involved in the cases. The learning curve seems plausible because the procedure has some nuances that get better with training and experience.

Despite the excellent procedural/in-hospital outcomes and durable S3/S3U performance, the rate of 1-year mortality was 13.4%. To explore this further, we

performed a multivariable analysis of the procedural and demographic factors that could predict 1-year mortality rate. This suggested that 1-year all-cause mortality after transseptal MViV was primarily associated with the patient's clinical presentation (cardiogenic shock) and preprocedural comorbidities such as ESRD, immunocompromised status, home oxygen, age, and glomerular filtration rate. In particular, ESRD was independently associated with 2.5× the hazards of 1-year mortality rates, similar to preprocedure cardiogenic shock. Thus, managing the comorbidities after a successful procedure may help reduce the mortality rate further.

Although all patients were considered high-risk for surgery by the sites, some patients were classified as low-risk and intermediate-risk on the basis of their STS score. It is probable that these patients had high-risk features that were not captured by the STS score. Because of the large sample size, we were able to perform further analysis stratified by STS risk score. The low-risk group, defined as those with an STS score <4, were younger and had a very low in-hospital mortality rate of 0.4% and a 1-year mortality rate of 4.9%. Similarly, the intermediate-risk group had significantly lower short-term and long-term mortality rates than the high-risk group. Recent 5-year follow-up data from the MITRAL trial (Mitral Implantation of Transcatheter Valves) reported durable transcatheter heart valve function after MViV with stable gradients and MR grade.[17] As the outcomes of MViV are excellent and procedural results are predictable and durable, MViV may be considered for intermediate surgical-risk patients after heart team discussion. In younger low surgical-risk patients who may be candidates for redo surgery, further data are required to compare long-term outcomes between the 2 strategies because durability data beyond 5 years are lacking.

Limitations

This study has several limitations that are associated with observational data. The TVT Registry data are self-reported by the sites, so there is some missing information. The STS/American College of Cardiology frequently audits the data to maintain accuracy, as has been described previously.[7] The registry only includes data on patients who underwent the procedure. If a patient did not undergo MViV for clinical or anatomic reasons, that information is not available. Because it is not mandatory to report MViV data to the TVT Registry, this analysis does not account for all patients undergoing MViV in the United States. The registry does not include data on postprocedure hemodynamics such as change in pulmonary artery pressure or pulmonary capillary wedge pressure, which could affect short-term and long-term outcomes. There were limited data on bioprosthetic valve fracture at the time of this analysis, so it was not included.

Conclusions

This large, multicenter, prospective, observational study confirms the safety and efficacy of transseptal MViV with S3 and S3U in the contemporary era. Although MViV is currently approved only for high surgical-risk patients, excellent outcomes were noted in low-risk and intermediate-risk groups as well. The overall volumes of transseptal MViV procedures and sites have increased significantly in the United States, with a trend toward reduced mortality rates in the contemporary era, but the median volume/site remains low. There may be an initial learning curve, which warrants appropriate procedural training, education, and sharing of best practices. With growing evidence, MViV may be considered for the majority of anatomically appropriate patients with degenerated bioprosthetic mitral valves after a heart team discussion. Future studies should evaluate long-term clinical outcomes.

ARTICLE INFORMATION

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Supplemental Material

Expanded Methods

Tables S1–S7

Figures S1–S5

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